
Beyond Correlation: Evaluating Structured Prompting and Neuro-Symbolic Parsing for LLM Causal Reasoning

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Abstract

Large Language Models (LLMs) often struggle to distinguish genuine causal relationships from mere correlations (Kiciman et al., 2024; Akiba, 2026; Jin et al., 2024). In this paper, we evaluate whether guiding LLMs to explicitly construct structured causal expressions improves their causal reasoning capabilities. We investigate this through two distinct paradigms: (1) Confounder-Aware Prompting (CAP), a pure-prompting pipeline that forces LLMs to explicitly articulate causal pathways and confounders in natural language, and (2) a Neuro-Symbolic Pipeline (NSP) that uses LLMs to parse queries into structured causal graphs, verifies them, and applies a deterministic symbolic solver. Evaluating across a suite of LLMs (e.g., DeepSeek, GLM, Qwen) on the CLadder benchmark, we find that while structural prompting (CAP) matches standard Chain-of-Thought (CoT) in accuracy and improves interpretability, standard heuristics in strong models often saturate zero-shot accuracy. Meanwhile, the neuro-symbolic approach (NSP) offers strict verification of rule-based causal inference for successfully parsed queries. Our findings reveal the trade-offs between internalized reasoning heuristics and explicit symbolic logic in modern LLMs.

1. Introduction

Causal reasoning remains a fundamental challenge for artificial intelligence. Large Language Models (LLMs) exhibit remarkable capabilities in natural language understanding, yet they often fall victim to the classic fallacy of conflating correlation with causation (Akiba, 2026). Recent studies suggest that generating intermediate reasoning steps,

such as Chain-of-Thought (CoT), enhances performance across various logical tasks (Han Lin, 2025; Yuzhe Wang, 2025; Ma et al., 2025). However, it remains underexplored whether forcing an explicitly structured causal extraction—either within the text generation itself or through an external parser—yields superior causal reasoning.

In this work, we propose to systematically evaluate explicitly structured causal generation, addressing the limitations of semantic matching (Yuzhe Wang, 2025) and unstructured graph encoding (Sheth et al., 2025). We introduce **Confounder-Aware Prompting (CAP)**, an advanced prompting technique, alongside a robust **Neuro-Symbolic Pipeline (NSP)** utilizing parsing and symbolic solving. We evaluate these approaches comprehensively against Direct Prompting and Standard CoT across multiple LLM families (e.g., DeepSeek, GLM, Qwen).

2. Methodology

To isolate the benefits of explicit causal structure generation and compare continuous language reasoning to discrete logic, we present two primary frameworks:

2.1. Framework I: Prompt-Based Structural Reasoning

We compare three core prompting strategies that rely entirely on the model’s internal latent reasoning:

- **Direct Prompting:** Directly asking the model if a causal claim is valid or invalid based on the given premise.
- **Standard Chain-of-Thought (CoT):** Appending “Let’s think step by step” to elicit unstructured reasoning traces before concluding.
- **Confounder-Aware Prompting (CAP):** A structured three-step process forcing the LLM through: (1) *Pathway Analysis*, (2) *Alternative Structure (Confounder) Identification*, and (3) *Deconfounded Judgment*.

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2.2. Framework II: Neuro-Symbolic Causal Pipeline (NSP)

While CAP operates within natural language, our second method decouples structure extraction from logic execution to ensure strict causal rule adherence:

1. **Parser:** The LLM acts solely as an information extractor, mapping the natural-language question into a structured causal JSON representation.
2. **Verifier:** A programmed verifier checks schema compliance, graph connectivity, and probability structures.
3. **Symbolic Solver:** A deterministic engine applies hard causal rules (e.g., backdoor adjustment, collider bias, average treatment effect calculation) onto the verified graph to produce the final analytical answer.

We provide a comparative evolution of the prompting strategies from unstructured to structured in Appendix A and an end-to-end NSP trace is provided in Appendix C.

3. Experimental Setup

Dataset and Models: We utilize the CLadder benchmark for complex causal relations (Jin et al., 2024; Yuzhe Wang, 2025). To ensure robust findings, we benchmark across multiple instruction-tuned API models including deepseek-v4-flash, LongCat-Flash-Chat, Qwen3.6, GLM-5, and MiniMax-2.7.

Evaluation Protocol: We measure overall accuracy (with Bootstrap 95% CI) and False Positive Rate (FPR). We conduct pairwise McNemar’s Tests with Bonferroni correction for rigorous method comparison.

4. Results

4.1. Main Performance Across Models

Our evaluation across multiple models (Table 1) demonstrates that LLMs fundamentally struggle with zero-shot validation without reasoning traces (e.g., DeepSeek-Chat Direct Prompting yields 58.65% accuracy).

Structurally guiding models (Std CoT and CAP) provides massive gains. For DeepSeek-Chat, CAP performs competitively at 91.23%, while Std CoT achieves 92.48%. A similar relative performance stabilization is consistently observed across the evaluated models.

4.2. Statistical Analysis

Pairwise McNemar’s tests indicate that both CAP and Standard CoT significantly outperform Direct Prompting

($p \approx 6.26 \times 10^{-46}$ for DeepSeek). The difference between structural prompting (CAP) and unstructured prompting (CoT) is surprisingly not significant ($p > 0.05$).

To visualize the statistical significance across all method pairs, we provide a heatmap of the McNemar’s test p -values in Figure 2.

4.3. Performance by Causal Rung (Stratified Analysis)

To comprehensively evaluate causal capabilities, we stratify model performance according to Judea Pearl’s Ladder of Causation (Pearl & Mackenzie, 2018): Association (Rung 1), Intervention (Rung 2), and Counterfactuals (Rung 3). Table 2 presents the breakdown for a subset of evaluated models. As typically observed in causal evaluations, models exhibit varying degradation as queries transition from observational distributions to complex counterfactuals. Furthermore, because the NSP solver currently does not support the logic required for Rung 3 counterfactuals (solved count is 0), its accurate resolution on this rung is impossible.

4.4. Structural Verifiability

While CAP does not necessarily surpass unstructured CoT in generic scalar accuracy, it provides structural determinism. The Neuro-symbolic Parser pipeline ensures 100% adherence to rule-based logic (zero causal hallucinations) for properly parsed graphs, addressing an interpretability gap that pure language CoT cannot bridge. We emphasize that the reported NSP accuracy is *solved-only*, as it does not guarantee a solution for every query if the LLM output is structurally invalid. In a pure language setup, CAP successfully articulated explicit confounder nodes in 78.45% of instances.

5. Conclusion

We explored explicitly exposing causal structures in LLM inference via two paths: Confounder-Aware Prompting (CAP) and a Neuro-Symbolic Parser-Solver. Both methods effectively cure the fatal zero-shot correlation-causation fallacies seen in Direct Prompting. However, we found that forcing explicit natural language causal graphs (CAP) performs statistically comparably to generic Chain-of-Thought across models, underscoring that while explicit constraints offer strict interpretability and parseable symbolic logic, modern LLMs possess internal reasoning heuristics adequately triggered by standard unstructured prompts.

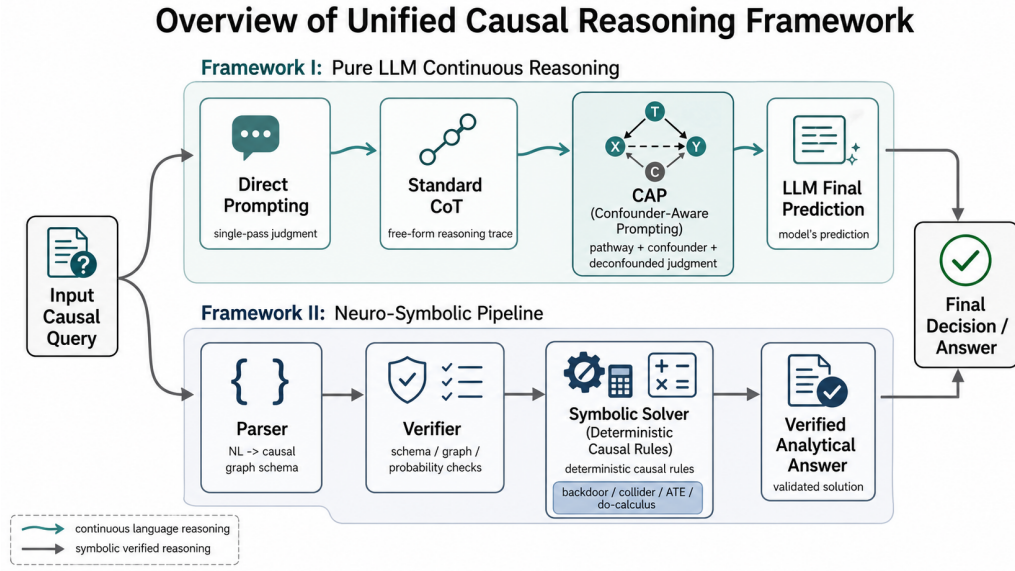


Figure 1. Overview of our unified causal reasoning framework. **Top Branch (Framework I):** Pure LLM continuous reasoning pathways including Direct Prompting, Standard CoT, and our proposed CAP. **Bottom Branch (Framework II):** The Neuro-Symbolic Pipeline that parses input into JSON graphs, verifies them, and uses a Symbolic Solver for deterministic inference.

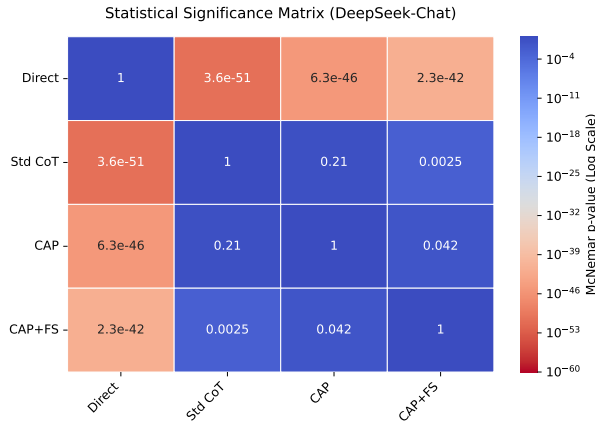


Figure 2. Significance matrix (McNemar’s test p -values) comparing the four prompting strategies based on the DeepSeek-Chat model results. Cells with warmer colors indicate statistically significant differences between methods ($p < 0.05$).

Model	Method	Acc (%)	95% CI	FPR (%)
DeepSeek-Chat	Direct	58.65	55.26-62.16	11.03
	Std CoT	92.48	90.60-94.24	7.27
	CAP	91.23	89.22-93.11	8.77
	CAP + FS	89.22	86.97-91.35	8.77
	NSP	-	-	-
DeepSeek-V4-Flash	Direct	88.47	86.22-90.73	9.52
	Std CoT	84.21	81.70-86.72	13.28
	CAP	80.58	77.82-83.33	17.29
	CAP + FS	83.08	80.45-85.71	15.79
	NSP	95.32	-	-
LongCat-Flash-Chat	Direct	65.54	62.16-68.80	19.30
	Std CoT	89.47	87.34-91.60	10.03
	CAP	89.72	87.59-91.85	6.52
	CAP + FS	87.59	85.21-89.85	8.77
	NSP	95.32	-	-
LongCat-Flash-Lite	Direct	58.54	55.20-61.88	47.14
	Std CoT	85.29	82.89-87.69	16.67
	CAP	87.44	85.19-89.69	11.67
	CAP + FS	85.41	83.01-87.80	15.00
	NSP	-	-	-
Qwen3.6	Direct	94.17	89.97-98.36	3.85
	Std CoT	92.44	87.69-97.19	3.85
	CAP	91.60	86.61-96.58	3.85
	CAP + FS	94.12	89.89-98.35	3.85
	NSP	-	-	-
GLM-5	Direct	83.33	79.63-87.03	13.16
	Std CoT	77.69	73.56-81.82	18.95
	CAP	81.49	77.63-85.35	16.40
	CAP + FS	81.75	77.91-85.59	16.93
	NSP	-	-	-
MiniMax-2.7	Direct	-	-	-
	Std CoT	-	-	-
	CAP	-	-	-
	CAP + FS	-	-	-
	NSP	-	-	-

Table 1. Overall Performance Metrics across multiple LLMs ($N = 798$). Data shown illustrates differing prompting paradigms. Note that the accuracy for the Neuro-Symbolic Pipeline (NSP) is calculated on a *solved-only* basis, as the solver may reject invalid graphs.

Model	Method	Rung 1	Rung 2	Rung 3
DeepSeek-Chat	Direct	59.02	59.77	57.14
	Std CoT	98.50	93.98	84.96
	CAP	95.49	96.24	81.95
	CAP + FS	97.00	89.10	81.58
	NSP	-	-	-
DeepSeek-V4-Flash	Direct	98.50	90.23	76.69
	Std CoT	98.87	87.59	66.17
	CAP	97.37	86.84	57.51
	CAP + FS	98.50	90.98	59.77
	NSP	97.89	92.70	-
LongCat-Flash-Chat	Direct	66.54	64.66	65.41
	Std CoT	95.11	90.60	82.71
	CAP	91.35	94.36	83.46
	CAP + FS	91.35	90.23	81.20
	NSP	97.89	92.70	-
LongCat-Flash-Lite	Direct	-	-	-
	Std CoT	-	-	-
	CAP	-	-	-
	CAP + FS	-	-	-
	NSP	-	-	-
Qwen3.6	Direct	-	-	-
	Std CoT	-	-	-
	CAP	-	-	-
	CAP + FS	-	-	-
	NSP	-	-	-
GLM-5	Direct	-	-	-
	Std CoT	-	-	-
	CAP	-	-	-
	CAP + FS	-	-	-
	NSP	-	-	-
MiniMax-2.7	Direct	-	-	-
	Std CoT	-	-	-
	CAP	-	-	-
	CAP + FS	-	-	-
	NSP	-	-	-

Table 2. Accuracy (%) stratified by Causal Rung. Rung 1 corresponds to Association, Rung 2 to Intervention, and Rung 3 to Counterfactuals. (NSP entries for Rung 3 are omitted due to the solver’s incompatibility with counterfactual queries).

References

- Akiba, D. The illusion of causality in llms: A developmentally grounded analysis of semantic scaffolding and benchmark-capability mismatches. *Machine Learning and Knowledge Extraction*, 8(3):57, 2026.
- Han Lin, M. J. Structured causal cot prompting for enhanced causal reasoning in llms. *Current Science*, 2025.
- Jin, Z., Liu, J., Lyu, Z., Peng, S., Berber Sardinha, T., Bottou, L., Schölkopf, B., and Sachan, M. Can large language models infer causation from correlation? In *The Twelfth International Conference on Learning Representations*, 2024.
- Kiciman, E., Ness, R., Sharma, A., and Tan, C. Causal reasoning and large language models: Opening a new frontier for causality. *arXiv preprint arXiv:2305.00050*, 2024.
- Ma, J. et al. Causal inference with large language model: A survey. In *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025.
- Pearl, J. and Mackenzie, D. *The Book of Why: The New Science of Cause and Effect*. Basic books, 2018.
- Sheth, I., Fatemi, B., et al. Causalgraph2llm: Evaluating llms for causal queries. In *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025.
- Yuzhe Wang, Yaochen Zhu, J. L. Causalfip: A benchmark for llm causal judgment beyond semantic matching. *arXiv preprint arXiv:2602.20094*, 2025.

A. Evolution of Prompting Strategies

To illustrate the increasing structural complexity and reasoning depth, we present the three prompting paradigms using the `cladder_4816` case.

Strategy 1: Direct	Strategy 2: Standard CoT	Strategy 3: CAP (Ours)
"...is it more correct to use the Method 1 than Method 2? Answer with exactly one word: Yes or No"	"...Let's think step by step about this causal question. Please provide your reasoning..."	"...Please analyze this question using the following three steps: Step 1... Step 2... Step 3..."
Model Output (DeepSeek-Chat): Yes	Model Output (DeepSeek-Chat): The causal graph is Husband $\rightarrow$ Wife $\rightarrow$ Alarm clock, and Husband $\rightarrow$ Alarm clock. The question asks how husband affects alarm clock... Since there are no unmentioned confounders, the general correlation reflects the total causal effect... Final answer: Yes	Model Output (DeepSeek-Chat): Step 1 — Identify the Causal Level: This is a Rung 2 causal question... Wife is a mediator between Husband and Alarm clock... Step 2 — Select the Identification Strategy: To estimate the total effect... we should not condition on Wife... Step 3 — Apply and Conclude: Method 1 is more appropriate... Final answer: Yes

Table 3. Comparison of prompting complexity and reasoning traces for Example `cladder_4816`.

B. Detailed Prompting Templates

B.1. CAP Prompt Template

All causal reasoning queries in the CAP framework follow this parameterized template:

```
Imagine a self-contained, hypothetical world with only the following conditions, and
without any unmentioned factors or causal relationships: {{Causal_Relationships}}.
Method 1: {{Description_of_Total_Effect}}. Method 2: {{
Description_of_Conditional_Effect}}. To understand how {{Treatment}} affects {{Outcome
}}, is it more correct to use the Method 1 than Method 2?

Please analyze this question using the following three steps:

**Step 1 - Identify the Causal Level:**
Is this question asking about (a) a statistical association (Rung 1), (b) the effect of an
intervention / do-operator (Rung 2), or (c) a counterfactual / hypothetical outcome (
Rung 3)? Identify the relevant variables and the causal graph structure.

**Step 2 - Select the Identification Strategy:**
Based on the causal level and graph structure, what is the correct method to answer this
question? For example: direct observation, backdoor adjustment, front-door criterion,
instrumental variable, or counterfactual reasoning. Are there any confounders or
colliders that could bias a naive analysis?

**Step 3 - Apply and Conclude:**
Apply the identified strategy to the given probabilities or causal structure. Does the
evidence support a 'Yes' or 'No' answer?

Final answer: Yes or No
```

B.2. Example Instantiation (`cladder_4816`)

Following the template in Appendix B.1, the instantiation for query ID `cladder_4816` is as follows:

CAP Instantiation: cladder_4816

Prompt:

Imagine a self-contained, hypothetical world with only the following conditions... Husband has a direct effect on wife and alarm clock. Wife has a direct effect on alarm clock. Method 1: We look directly at how husband correlates with alarm clock in general. Method 2: We look at this correlation case by case according to wife. To understand how husband affects alarm clock, is it more correct to use the Method 1 than Method 2?

... (3 Steps prompt appended)

Model Response (Example):

Step 1 | Identify the Causal Level:

This is a Rung 2 causal question because it asks how Husband affects Alarm clock. The relevant causal graph is: Husband → Wife → Alarm clock

Husband → Alarm clock

So Wife is a mediator between Husband and Alarm clock, not a confounder.

Step 2 | Select the Identification Strategy:

To estimate the total effect of Husband on Alarm clock, we should not condition on Wife. Conditioning on Wife would block the mediated path Husband → Wife → Alarm clock, leaving only the direct effect Husband → Alarm clock. Therefore, no backdoor adjustment is needed.

Step 3 | Apply and Conclude:

Because the goal is to understand the total effect of Husband on Alarm clock, Method 1 is more appropriate than Method 2.

Final answer: Yes

B.3. Few-Shot Learning Setup (CAP + FS)

To provide in-context learning, the CAP + Few-shot explicitly precedes the standard template with three solved examples representing each of Pearl's three causal rungs:

- **Example 1 (Rung 1 — Association):** Demonstrates marginal probability calculation without intervention (e.g., weather and ice cream sales).
- **Example 2 (Rung 2 — Intervention):** Demonstrates the application of the *do*-operator without backdoor confounding (e.g., exercise, diet, and fitness).
- **Example 3 (Rung 3 — Counterfactual):** Demonstrates counterfactual reasoning under assumptions of no unmeasured confounders (e.g., studying and exam scores).

By analyzing standard examples across the ladder of causation, the model learns the structured three-step reasoning format and adheres strictly to the desired analytical pathway.

C. NSP Case Study: Parser–Verifier–Solver

To mirror the presentation style of prompting examples, we provide a full Neuro-Symbolic Pipeline (NSP) walkthrough using cladder_22688.

C.1. Case Metadata and Query Type

Case Metadata: cladder_22688

ID: cladder_22688

Dataset: CLadder

Query type: marginal

Graph type: mediation

Rung: 1 (Association)

Ground truth: Yes

C.2. NSP Flow and Candidate Status

Candidate	Parser	Verifier	Note
0	Success	Invalid	Missing variables/edges/probability facts
1	Success	Valid	State naming warning (yes/no)
2	Success	Valid	State naming warning (jyka/not_jyka)

Table 4. Parser and verifier outcomes for three candidate parses.

C.3. Selected Parse (Aggregator Output)

Aggregator Decision

Status: plurality
 Selected candidate: 2
 Selected rule: law_of_total_probability
 Support: 1 / 2 valid candidates (agreement = 0.5)

Listing 1. Selected causal structure and probability facts

```
Variables:
- jyka (treatment), states: [jyka, not_jyka]
- hwax (mediator), states: [hwax, not_hwax]
- lirr (outcome), states: [lirr, not_lirr]

Edges:
jyka -> hwax
jyka -> lirr
hwax -> lirr

Facts:
P(jyka) = 0.41
P(lirr | jyka) = 0.50
P(lirr | not_jyka) = 0.44
```

C.4. Symbolic Solver Calculation and Final Answer

Law of Total Probability

$$P(lirr) = P(lirr | jyka)P(jyka) + P(lirr | not_jyka)P(not_jyka)$$

$$= 0.50 \times 0.41 + 0.44 \times 0.59 = 0.4646$$

Hence, $P(lirr) = 46.46\% < 50\%$, so $P(lirr) < P(not_lirr)$.

Final answer: Yes